

Exercise 1: Configuring a Basic Spring Application

Scenario:

Your company is developing a web application for managing a library. You need to use the Spring Framework to handle the backend operations.

```
J BookRepository.java 1  J BookService.java 6  applicationContext.xml  J LibraryManagementApplication.java 8 X  pom.xml
week-3 > src > main > java > com > library > J LibraryManagementApplication.java > LibraryManagementApplication
1  package com.library;
2
3  import org.springframework.context.ApplicationContext;
4  import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6  import com.library.service.BookService;
7
8  public class LibraryManagementApplication {
9
10     public static void main(String[] args) {
11
12         ApplicationContext context =
13             new ClassPathXmlApplicationContext("applicationContext.xml");
14
15         BookService service =
16             context.getBean("bookService", BookService.class);
17
18         service.showBooks();
19     }
20 }
```

```
week-3 > src > main > java > com > library > repository > J BookRepository.java > BookRepository
1  package com.library.repository;
2
3  public class BookRepository {
4
5      public void displayBooks() {
6          System.out.println("Fetching books from repository...");
7      }
8  }
```

```
J BookRepository.java 1  J BookService.java 6 X  applicationContext.xml  J LibraryManagementApplication.java 8  pom.xml
week-3 > src > main > java > com > library > service > J BookService.java > BookService
1  package com.library.service;
2
3  import com.library.repository.BookRepository;
4
5  public class BookService {
6
7      private BookRepository repository;
8
9      public void setRepository(BookRepository repository) {
10         this.repository = repository;
11     }
12
13     public void showBooks() {
14         System.out.println("Book Service Started");
15         repository.displayBooks();
16     }
17 }
```

O/P :

```
[INFO] Attaching agents: []
Book Service Started
Fetching books from repository...
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 16.876 s
[INFO] Finished at: 2026-07-02T16:25:25+05:30
[INFO]
```

Exercise 2: Implementing Dependency Injection

```
1 package com.library;
2 import org.springframework.context.ApplicationContext;
3 import org.springframework.context.support.ClassPathXmlApplicationContext;
4
5 import com.library.service.BookService;
6
7 public class MainApp {
8
9     public static void main(String[] args) {
10
11         ApplicationContext context =
12             new ClassPathXmlApplicationContext("applicationContext.xml");
13
14         BookService service =
15             context.getBean("bookService", BookService.class);
16
17         service.showBooks();
18
19     }
20
21 }
```

```
J BookRepository.java 1 J MainApp.java J BookService.java 6 X applicationContext.xml J LibraryManagementApplication.java 8 pom.xml 1
week-3 > src > main > java > com > library > service > J BookService.java

1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4
5 public class BookService {
6
7     private BookRepository repository;
8
9     public void setRepository(BookRepository repository) {
10         this.repository = repository;
11     }
12
13     public void showBooks() {
14         System.out.println("Dependency Injection Successful!");
15         repository.displayBooks();
16         System.out.println("Injected Object : " + repository);
17     }
18 }
```

O/P :

```
[INFO] -----< com.library:LibraryManagement >-----
[INFO] Building LibraryManagement 0.0.1-SNAPSHOT
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.6.3:java (default-cli) @ LibraryManagement ---
Dependency Injection Successful!
BookRepository bean invoked.
Retrieving library books...
Injected Object : com.library.repository.BookRepository@55b8724a
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time:  1.303 s
[INFO] Finished at: 2026-07-02T17:01:04+05:30
[INFO] -----
PS C:\Users\Kanna\OneDrive\Desktop\COGNIZANT\DEEPAK\WEEK1\WEEK_1_MANDATORY\week-3>
```

Exercise 4: Creating and Configuring a Maven Project

```
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4     https://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6   <modelVersion>4.0.0</modelVersion>
7
8   <groupId>com.library</groupId>
9   <artifactId>LibraryManagement</artifactId>
10  <version>1.0</version>
11  <packaging>jar</packaging>
12
13  <dependencies>
14
15    <dependency>
16      <groupId>org.springframework</groupId>
17      <artifactId>spring-aop</artifactId>
18      <version>6.2.7</version>
19    </dependency>
20
21    <dependency>
22      <groupId>org.springframework</groupId>
23      <artifactId>spring-context</artifactId>
24      <version>6.2.7</version>
25    </dependency>
26
27  </dependencies>
28
29  <build>
30
31    <plugins>
32
33      <plugin>
34        <groupId>org.apache.maven.plugins</groupId>
35        <artifactId>maven-compiler-plugin</artifactId>
36        <version>3.13.0</version>
37
38        <configuration>
39          <source>1.8</source>
40          <target>1.8</target>
41        </configuration>
42      </plugin>
43
44    </plugins>
45
46  </build>
47
48 </project>
```

O/P :

```
PS C:\Users\Kanna\OneDrive\Desktop\COGNIZANT\DEEPAK\WEEK1\WEEK_1_MANDATORY\week-3> mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.library:LibraryManagement >-----
[INFO] Building LibraryManagement 1.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ LibraryManagement ---
[INFO] Deleting C:\Users\Kanna\OneDrive\Desktop\COGNIZANT\DEEPAK\WEEK1\WEEK_1_MANDATORY\week-3\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ LibraryManagement ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 2 resources from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ LibraryManagement ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 4 source files with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
not setting the bootstrap class path may lead to class files that cannot run on JDK 8
--release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.063 s
[INFO] Finished at: 2026-07-02T17:17:00+05:30
[INFO]
PS C:\Users\Kanna\OneDrive\Desktop\COGNIZANT\DEEPAK\WEEK1\WEEK_1_MANDATORY\week-3>
```